

QSKILL INTERNSHIP
ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

IRIS FLOWER CLASSIFICATION

Machine Learning Classification Project

Internship Domain	Artificial Intelligence & Machine Learning
Project	Iris Flower Classification
Model Used	Logistic Regression
Tools	Python, Google Colab, Pandas, Matplotlib, Seaborn, Scikit-learn

Student Name: Yashwant Ravindra Sonawane

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1. Objective

The objective of this project is to classify Iris flowers into three species—Setosa, Versicolor, and Virginica—using their sepal and petal measurements. A Logistic Regression classification model is trained and evaluated on the Iris dataset.

2. Introduction

Iris Flower Classification is a supervised machine learning problem in which a model learns the relationship between flower measurements and their species. The project uses four numerical features: sepal length, sepal width, petal length, and petal width. The trained model is then used to predict the species of unseen flower measurements.

3. Dataset Description

The project loads the Iris dataset using scikit-learn's `load_iris()` function. The dataset contains 150 samples and four measurement features. A species column is created with the three target classes: setosa, versicolor, and virginica.

Dataset Property	Value
Total samples	150
Features	4
Classes	3
Samples per class	50 each
Missing values	0

4. Features Used

- Sepal Length (cm)
- Sepal Width (cm)
- Petal Length (cm)
- Petal Width (cm)

5. Tools and Technologies

- Python 3
- Google Colab
- Pandas
- Matplotlib
- Seaborn
- Scikit-learn

6. Methodology

- 1. Dataset Loading:** The Iris dataset is loaded using scikit-learn.
- 2. DataFrame Creation:** The feature data is converted into a Pandas DataFrame and the species labels are added.

- 3. Data Exploration:** Dataset shape, column names, class counts, and missing values are checked.
- 4. Visualization:** Scatter plots are created for sepal measurements and petal measurements.
- 5. Train-Test Split:** The data is divided into 80% training data (120 samples) and 20% testing data (30 samples).
- 6. Model Training:** A Logistic Regression model with max_iter=200 is trained on the training data.
- 7. Prediction:** The trained model predicts classes for the test data.
- 8. Evaluation:** Accuracy, precision, recall, F1-score, and a confusion matrix are used to evaluate the model.
- 9. New Sample Prediction:** A new flower with measurements [5.1, 3.5, 1.4, 0.2] is classified.

7. Data Exploration

The notebook reports a dataset shape of (150, 5) after adding the species column. There are 50 samples for each species, and all five columns contain zero missing values.

8. Data Visualization

Sepal measurements:

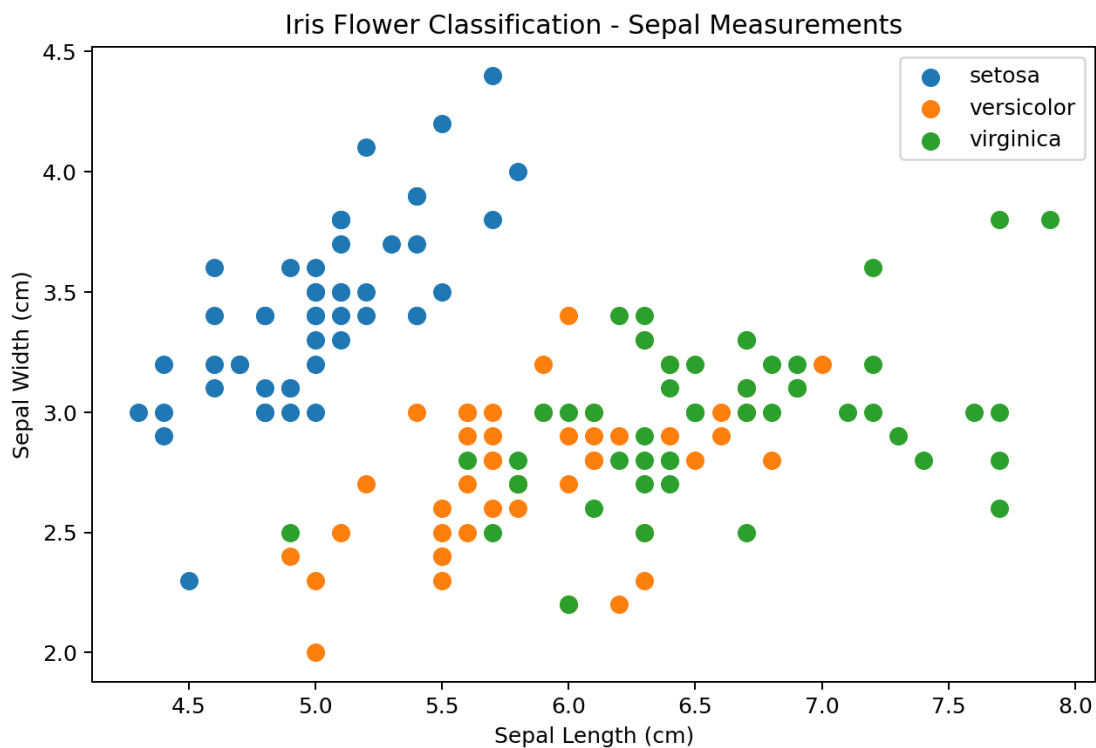


Figure 1: Scatter plot of sepal length and sepal width.

Petal measurements:

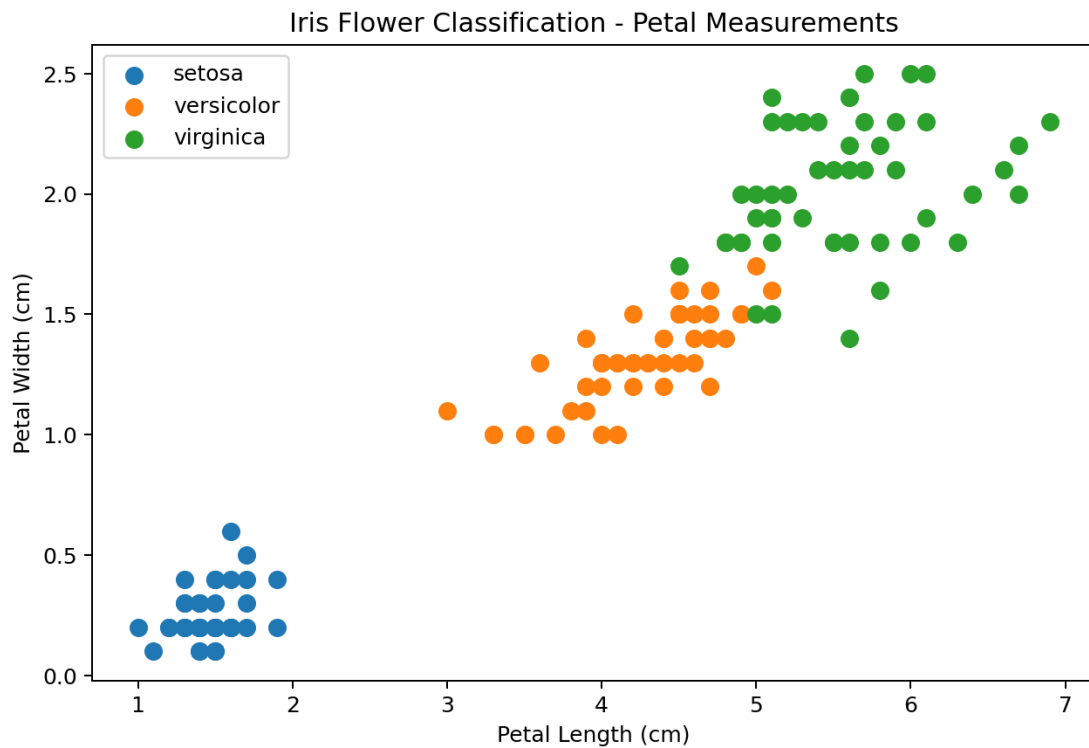


Figure 2: Scatter plot of petal length and petal width.

9. Model Training

The dataset is split using `train_test_split` with `test_size=0.2`, `random_state=42`, and `stratify=y`. This produces 120 training samples and 30 testing samples. Logistic Regression is then fitted using the training data.

10. Model Evaluation

The trained model achieved an accuracy of 96.67% on the 30-sample test set.

Class	Precision	Recall	F1-score	Support
setosa	1.00	1.00	1.00	10
versicolor	1.00	0.90	0.95	10
virginica	0.91	1.00	0.95	10
Accuracy			0.97	30

Confusion Matrix:

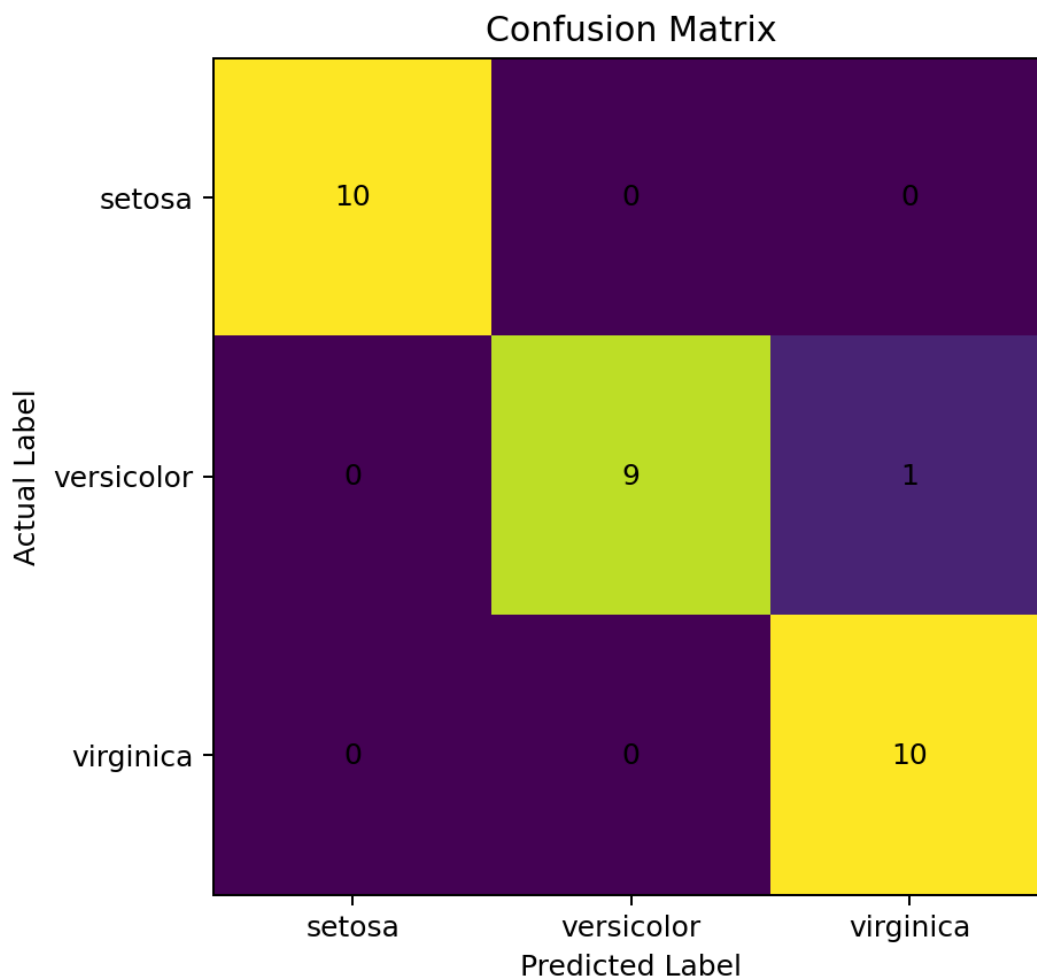


Figure 3: Confusion matrix for the test predictions.

11. New Flower Prediction

For the new flower measurements [5.1, 3.5, 1.4, 0.2], the trained model predicted:

SETOSA

12. Results

Result	Value
Dataset samples	150
Training samples	120
Testing samples	30
Model	Logistic Regression
Accuracy	96.67%
New sample prediction	Setosa

13. Conclusion

The Iris Flower Classification project was successfully implemented using Python and scikit-learn. The Logistic Regression model classified the three Iris species and achieved 96.67% accuracy on the test dataset. The classification report and confusion matrix provide additional evaluation of the model. The trained model also predicted the sample flower with measurements [5.1, 3.5, 1.4, 0.2] as Setosa.

14. Project Files

The Google Colab notebook contains the complete implementation, including dataset loading, exploration, visualization, model training, prediction, and evaluation.